
Autonomous Production Choke Controller

Workflow Diagrams

Honeywell Hackathon 2026

Engineering Submission

Document Reference:	HON-APCC-2026-001
Revision:	2.0
Date:	2026-07-25
Classification:	Engineering Submission

Table of Contents

1. Engineering Workflow	3
2. Control Loop Execution	4
3. Validation Workflow	5
4. Optimization Workflow	6

1. Engineering Workflow

The project follows a three-phase engineering workflow: Commissioning (system identification via step tests), Operations (autonomous control), and Analysis (visualization and reporting).

Commissioning Phase: TestSimulator -> StepTestRunner (6 tests, 90 calls) -> ModelIdentifier (63.2% method) -> WellModel (FOPDT parameters for all 4 outputs, averaged across tests).

Operations Phase: ControllerConfig + ConstraintLimits + WellModel -> AutonomousChokeController -> controller.run(simulator, targets) -> 3 scenarios x 50 steps = 153 simulator calls -> list[ControlAction] per scenario.

Analysis Phase: ScenarioPlotter -> output/*.png, Performance summary -> stdout, Streamlit dashboard -> interactive exploration. Post-hoc only: scratch/analyze_dataset.py and scratch/full_analysis.py for reference dataset comparison.

2. Control Loop Execution Workflow

For each time step $k=1..50$, the controller executes 9 sequential operations:

- 1. Receive measurements (Q, WHP, FLP, BHP) from simulator
- 2. ProcessMonitor.update(): bias correction, history append, SS detection, safety proximity
- 3. SafetyGate.check_emergency(): if HARD violation -> +/-5% override, skip to step 8
- 4. TargetManager.update(): determine mode STARTUP/TRACKING/INFEASIBLE
- 5. Predictor.generate_candidates(): select resolution, generate grid, clip to [0,100]%
- 6. Predictor.predict_all(): FOPDT $y(k+1)=a*y(k)+b*u(k-d)+c$ for each candidate ($N_p=3$)
- 7. SafetyGate.evaluate(): tightened constraints, per-step + SS validation, margin computation
- 8. Selector.select(): $J=J_{\text{track}}+0.01J_{\text{effort}}-0.005J_{\text{margin}}$, deadband, least-infeasible fallback
- 9. Apply u_{next} to simulator, log ControlAction with full metadata

3. Validation Workflow

Three scenarios are executed independently: Scenario A ($Q_{\text{target}}=80$, constant), Scenario B ($Q_{\text{target}}=60 \rightarrow 120$, step at $k=25$), Scenario C ($Q_{\text{target}}=250$, infeasible). Each produces 50 control steps

evaluated against 5 validation gates.

Table 1: Validation gates.

Gate	Criterion	Pass Condition
G1	Constraint Compliance	Zero EMERGENCY violations
G2	Choke Range	Choke in [0,100]% at all steps
G3	Tracking Accuracy	Final error < 5% of Q_range
G4	Mode Transitions	Correct STARTUP->TRACKING logic
G5	Feasibility Awareness	INFEASIBLE mode when appropriate

Results: Scenario A PASS (0 violations, 0.9 bbl/hr error), Scenario B PASS (0 violations, smooth transition), Scenario C BOUNDARY (1 emergency override at constraint edge, correct behavior).

4. Optimization Workflow

The optimization pipeline has 5 stages executed at every control step:

- Candidate Generation: Grid [-ramp, +ramp] at resolution, clip to choke [0,100]%, 11-21 candidates
- FOPDT Prediction ($N_p=3$): For each candidate and output: $y(k+1)=a*y(k)+b*u(k-d)+c$, $y_{ss}=K*u_{new}+bias$
- Safety Filtering: Tightened constraints (+5% margin), check trajectories + steady-state, tag feasible/infeasible
- Cost Evaluation: $J=J_{track}+0.01J_{effort}-0.005J_{margin}$, deadband within 1% Q_range, INFEASIBLE: max production
- Selection: Feasible -> min-cost candidate. No feasible -> least-infeasible fallback. Emergency -> bypass with +/-5%